

(1)

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Solutions

1.

$$\frac{dv}{dx} = \frac{dv}{du} \cdot \frac{du}{dx}$$

$$\text{when } x = 2$$

$$u = 4$$

$$= \left(\frac{(u-1) - u}{(u-1)^2} \right) \cdot (2x)$$

$$= \frac{-1}{(u-1)^2} (2x)$$

$$= \frac{-1}{9} (4)$$

$$= \frac{-4}{9}$$

2.

$$\frac{dv}{dx} = \frac{dv}{du} \cdot \frac{du}{dx}$$

$$= \frac{u^2 + 1 - 2u^2}{(u^2 + 1)^2}$$

$$\text{when } x = 1$$

$$u = 0$$

$$= \frac{-u^2 + 1}{(u^2 + 1)^2} (3x^2 - 1)$$

$$= 2$$

3. $f(x) = x^3 - 3x + 2$

$$f'(x) = 3x^2 - 3$$

$$3x^2 - 3 = 0$$

horizontal tangent; $m_t = 0$

$$x = \pm 1$$

$\therefore$ pts are $(1, 0)$ & $(-1, 4)$

4. $f(x) = g(x)h^3(x)$

$$f'(x) = g'(x)h^3(x) + 3g(x)h^2(x) \cdot h'(x)$$

$$f'(2) = g'(2)h^3(2) + 3g(2)h^2(2)h'(2)$$

$$f'(2) = (3)(-2)^3 + 3(-1)(-2)^2(4)$$

$$f'(2) = -24 - 48$$

$$f'(2) = -72$$

(3)

$$5. \quad x + 5y - 10 = 0$$

$$y = -\frac{1}{5}x + 2 \Rightarrow m_t = 5 (\perp)$$

$$y = 3x^2 - 7x + 5$$

$$y' = 6x - 7$$

$$6x - 7 = 5$$

$$x = 2$$

$$y - y_1 = m(x - x_1)$$

$$y - 3 = 5(x - 2)$$

$$6. \quad f(x) = 3 + 2x + 3x^2$$

$$f'(x) = 6x + 2$$

$$y - y_1 = m(x - x_1)$$

$$3 + 2x + 3x^2 + 23 = (6x + 2)(x + 1)$$

$$3x^2 + 2x + 26 = 6x^2 + 8x + 2$$

$$-3x^2 - 6x + 24 = 0$$

$$x^2 + 2x - 8 = 0$$

$$(x+4)(x-2) = 0$$

pts of tangency
are $(-4, 43)$ &
 $(2, 19)$

$$* \therefore y - 43 = -22(x + 4)$$

$$y - 19 = 14(x - 2)$$